

A Deformed Anticommuting M_4 Counterexample

agent_lane_03

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Source and Conjecture

The source paper is David P. Blecher, *Commutativity of operator algebras*, arXiv:2509.17673. In Example 5.2, the paper constructs the noncommutative reversible algebra

$$A_1 = \text{span}\{e_{13} + e_{24}, e_{12} - e_{34}, e_{14}\} \subset M_4.$$

Immediately after this example, on page 13, it states the following conjecture:

“We conjecture that up to unitary equivalence A is the only noncommutative reversible 4×4 matrix algebra.”

We can also verify the conjecture z, w from Example 4.7 in this case. Indeed $P(x) = pxq$ is the canonical projection onto $Z = I(A)$, and so $z = P(1) = pq$. And $w = -z = -pq$ in this example. So $z \neq w$.

Finally we remark that all the matrices considered in this example have real entries, and so the example has the same conclusions in the real case. We conjecture that up to unitary equivalence A is the only noncommutative reversible 4×4 matrix algebra. Again this may be a nice undergraduate research project.

Corollary 5.3. *If A is a symmetric operator algebra then its unitization A^1 need not be symmetric. If A is a reversible operator algebra then A^1 need not be a*

Figure 1: The source conjecture in arXiv:2509.17673, p. 13.

The conjecture is false as stated with unitary equivalence.

Definitions

For $1 \leq i, j \leq 4$, let e_{ij} denote the standard matrix unit in M_4 . A concrete operator algebra $A \subset M_4$ is *reversible* if the same operator space, equipped with the reversed product

$$x \circ y = yx,$$

is again an abstract operator algebra.

Proof Intuition

The source example is anticommuting: every product reverses sign when the factors are interchanged. Anticommutation is stronger than reversibility, because the map $x \mapsto -x$ identifies the reversed product with the original product completely isometrically.

The point is that the source CAR-normalized matrices are not the only anticommuting realization in M_4 . One can scale one leg of the construction and still keep $UV = -VU$. The resulting algebra remains reversible and noncommutative, but a simple unitary invariant changes: after the unique rank-one product line is fixed, each non-product element has a fixed ratio between its “lower” $e_4 \rightarrow \text{span}\{e_2, e_3\}$ component and its $\text{span}\{e_2, e_3\} \rightarrow e_1$ component. The source example has ratio 1; the example below has ratio 2.

The Counterexample

Theorem 1. *Let*

$$U = e_{13} + 2e_{24}, \quad V = e_{12} - 2e_{34}, \quad W = e_{14},$$

and put

$$A_2 = \text{span}\{U, V, W\} \subset M_4.$$

Then A_2 is a noncommutative reversible 4×4 matrix algebra, but A_2 is not unitarily equivalent to the source algebra A_1 .

Proof. The matrix-unit relations give

$$U^2 = V^2 = UW = WU = VW = WV = 0,$$

and

$$UV = -2W, \quad VU = 2W.$$

Hence, for all $x, y \in A_2$,

$$xy = -yx.$$

In particular A_2 is not commutative, since $VU = 2W \neq 0$.

The anticommutation identity also proves reversibility. Indeed, in the reversed algebra A_2^{rev} the product is

$$x \circ y = yx = -xy.$$

The map $\Phi : A_2^{\text{rev}} \rightarrow A_2$, $\Phi(x) = -x$, is a completely isometric algebra isomorphism, since

$$\Phi(x \circ y) = xy = \Phi(x)\Phi(y).$$

Thus A_2^{rev} is an operator algebra.

It remains to prove that A_2 is not unitarily equivalent to A_1 . More generally, for $t > 0$ set

$$U_t = e_{13} + te_{24}, \quad V_t = e_{12} - te_{34}, \quad A_t = \text{span}\{U_t, V_t, W\}.$$

We show that A_t and A_s are unitarily equivalent only if $t = s$.

First note that $A_t^2 = \mathbb{C}W$. Moreover the nonzero rank-one elements of A_t are exactly the nonzero scalar multiples of W . Indeed an arbitrary element

$$X = aU_t + bV_t + cW$$

has matrix

$$X = \begin{bmatrix} 0 & b & a & c \\ 0 & 0 & 0 & at \\ 0 & 0 & 0 & -bt \\ 0 & 0 & 0 & 0 \end{bmatrix}.$$

If $(a, b) \neq (0, 0)$, then the range of X contains a nonzero vector in $\mathbb{C}e_1$ and also the nonzero vector $t(ae_2 - be_3)$ modulo $\mathbb{C}e_1$; so $\text{rank } X = 2$. If $a = b = 0$, then $X = cW$ has rank at most one.

Suppose that a unitary Q satisfies $QA_tQ^* = A_s$. Since unitary conjugacy preserves rank, it sends the unique rank-one line $\mathbb{C}W$ onto itself. Therefore $Qe_1 \in \mathbb{C}e_1$ and $Qe_4 \in \mathbb{C}e_4$, and hence

$$E = \text{span}\{e_2, e_3\}$$

is invariant under Q .

For $X \in A_t \setminus \mathbb{C}W$, define

$$r(X) = \frac{\|P_E X e_4\|}{\|P_{\mathbb{C}e_1} X|_E\|},$$

where P_E and $P_{\mathbb{C}e_1}$ are the orthogonal projections onto E and $\mathbb{C}e_1$, respectively. This ratio is unchanged by such a unitary conjugacy, because Q preserves the three orthogonal summands $\mathbb{C}e_1$, E , and $\mathbb{C}e_4$ up to unitary maps.

For $X = aU_t + bV_t + cW$ with $(a, b) \neq (0, 0)$,

$$\|P_E X e_4\| = t\sqrt{|a|^2 + |b|^2}, \quad \|P_{\mathbb{C}e_1} X|_E\| = \sqrt{|a|^2 + |b|^2}.$$

Thus $r(X) = t$ for every $X \in A_t \setminus \mathbb{C}W$. Similarly every non-product element of A_s has ratio s . Therefore unitary equivalence forces $t = s$. Taking $t = 2$ and $s = 1$, A_2 is not unitarily equivalent to the source algebra A_1 . $\square$

Verification Notes

The script `code/check_deformed_anticommuting.py` verifies the displayed matrix identities for $t = 2$, checks that $U + V$ has rank 2, and evaluates the invariant ratio on several sample elements. It is only a sanity check; the proof above is exact.

Command used:

```
conda run --no-capture-output -n sandbox python
code/check_deformed_anticommuting.py
```

The output gives zero residuals for $U^2, V^2, UV + 2W, VU - 2W, UV + VU$, singular values $(1, 1, 0, 0)$ for the source U_1 , singular values $(2, 1, 0, 0)$ for the deformed U_2 , and ratio 2 in the tested non-product elements.

Scope and Novelty Check

This packet refutes the source conjecture as literally stated with unitary equivalence. It does not classify all reversible 4×4 matrix algebras, and it does not address whether the deformed examples may be equivalent under some weaker nonunitary similarity or completely isometric algebraic equivalence.

The run's cheap indexes were searched for arXiv:2509.17673, the source title, “commutativity of operator algebras”, “reversible operator algebra”, “anticommuting operator algebra”, and “ 4×4

matrix algebra”. No prior packet or attempt for this source conjecture was found. A bounded web search for the exact conjecture phrase and nearby terms found the source preprint and unrelated uses of anticommutation/reversibility, but no existing counterexample to this unitary-equivalence uniqueness conjecture.

Human Review Recommendation

Recommended verdict: likely valid counterexample to the unitary-equivalence conjecture in Example 5.2. Reviewers should focus on the last invariant argument: unitary conjugacy must preserve the unique rank-one line $\mathbb{C}e_{14}$, hence the decomposition $\mathbb{C}e_1 \oplus \text{span}\{e_2, e_3\} \oplus \mathbb{C}e_4$, and therefore the ratio $r(X)$.

References

- [1] David P. Blecher, *Commutativity of operator algebras*, arXiv:2509.17673, 2025.